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Older adults' perceptions of technologies aimed at falls prevention, detection or monitoring: A systematic review

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ABSTRACT

Background: Over recent years a number of Information and Communication Technologies (ICTs) have emerged aiming at falls prevention, falls detection and alarms for use in case of fall. There are also a range of ICT interventions, which have been created or adapted to be pro-active in preventing falls, such as those which provide strength and balance training to older adults in the prevention of falls. However, there are issues related to the adoption and continued use of these technologies by older adults.

Objectives: This review provides an overview of older adults' perceptions of falls technologies.

Methods: We undertook systematic searches of MEDLINE, EMBASE, CINAHL and PsychINFO, COMPENDEX and the Cochrane database. Key search terms included 'older adults', 'seniors', 'preference', 'attitudes' and a wide range of technologies, they also included the key word 'fall*'. We considered all studies that included older adults aged 50 and above. Studies had to include technologies related specifically to falls prevention, detection or monitoring. The Joanna Briggs Institute (JBI) tool and the Quality Assessment Tool for Quantitative Studies by the Effective Public Health Practice Project (EPHPP) were used.

Results: We identified 76 potentially relevant papers. Some 21 studies were considered for quality review. Twelve qualitative studies, three quantitative studies and 6 mixed methods studies were included. The literature related to technologies aimed at predicting, monitoring and preventing falls suggest that intrinsic factors related to older adults' attitudes around control, independence and perceived need/requirements for safety are important for their motivation to use and continue using technologies. Extrinsic factors such as usability, feedback gained and costs are important elements which support these attitudes and perceptions.

Conclusion: Positive messages about the benefits of falls technologies for promoting healthy active ageing and independence are critical, as is ensuring that the technologies are simple, reliable and effective and tailored to individual need. The technologies need to be clearly described in research and older peoples' attitudes towards different sorts of technologies must be clarified if specific recommendations are to be made.

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1. Introduction

Falls are an important public health issue. Each year, 35% of over-65s experience one or more falls. About 45% of people aged over 80 who live in the community fall each year. Between 10 and 25% of such fallers will sustain a serious injury [1]. This has implications in terms of independence, quality of life and also cost to health services [1]. Hip fracture is the most common serious injury related to falls in older people and death rates are continuing to rise [2]. Each year approximately 10% of the elderly population (65+) will be treated by a doctor for an injury and approximately 100,000 older people in the EU27 (countries within the European Union) and European Economic Area countries will die from injury from a fall [3].

Over recent years a number of Information and Communication Technologies (ICTs) have emerged aimed at falls prevention, fall detection and alarms for use in case of fall. There are a range of ICT interventions that have been adopted to monitor falls and alert professionals or carers [4] if a fall occurs, including home automation systems. To date these technologies tend to be re-active, which help to reduce a long lie (lying on the floor for long periods of time can cause serious health complications) and allow help to be brought quickly to the person who has fallen. There are also a range of ICT interventions which have been created or adapted to be pro-active in preventing falls, such as those which provide strength and balance training to older adults in the prevention of falls e.g. exergames, Wii-fit, Kinect [5,6]. There is increasingly strong evidence that exercise programmes that include specific strength and balance exercises can significantly reduce the risk and rate of falls [7,8]. Therefore, ICT innovations that can deliver exercise programmes in the home have the potential to reduce costs and increase adherence by providing regular feedback and motivation to older adults.

One of the main issues with use of ICT devices in the home is related to adoption and use of the systems. We know very little about older adults' attitudes towards falls interventions that use technologies. There is some general evidence around a range of intrinsic and extrinsic factors which motivate older adults to adopt the use of ICT devices [9,10]. However, work has not been done to examine the specific evidence related to falls interventions. Maintaining independence has been previously cited as a key reason for why people engage in falls prevention activities and also home exercise [11–13]. Independence has also been found to be a key factor in promoting older adults' adoption of general ICT devices in their homes. This is often because they do not want to be a burden, have a fear of being put in long-term care, or because they feel it will promote their social life [9,14].

This review aims to provide guidance related to the attractiveness of specific ICT interventions and monitoring equipment, which is directly related to falls prevention and detection. Previous reviews of ICT [15] found that there is a lack of eligible randomised controlled trials (RCTs) in this area. As part of a broader European project (farseeingresearch.eu) we intend to draw together the literature that does exist (RCT, case studies, cohort studies, quasi experiments, qualitative and other methods) to provide a summary of existing knowledge and to identify gaps for future research.

Table 1 – Search terms for ICT interventions/applications/technologies.

Activity monitoring	Intelligent environment
Accelerometer	Interactive multimedia
Alarm button	Interactive technology
Alert system	Internet
Ambient assisted living	Internet-based intervention
Assistive technology	Keyboard
Audio visual	Message
Camera	Microphone
CD-ROM	Mobile
Cell phone	Monitoring
Cell-phone	Mouse
Cellular phone	Notebook
Communication network$	Personal digital assistant
Computer	PDA
Computerised	Persuasive technology
Computerised reminder$	Phone-based intervention
Databases	Reminder system
Detection	Remote consultation$
Electronic	Smart phone
Email	Smart Technology
E-mail	Tablet
eHealth	Technology
Fall alarm	Telecare
Fall detector	Telehealth
Gyroscope	Telemedicine
Headphone	Telenursing
Headset	Tele-rehabilitation
Home alarm	Touchscreen
Home automation	Video
Home nursing	Video-phone intervention
Home care	Virtual
Home-telecare intervention	Web
Home interactive	Website
ICT	Wireless
Informatics	World wide web
Information Services	www

2. Methods

2.1. *Search strategy*

We undertook systematic searches of MEDLINE, EMBASE, CINAHL and PsychINFO. We also undertook a systematic search of the engineering database COMPENDEX and the Cochrane database. No date restrictions were placed on the searches and all relevant evidence was included if in the English language. Search terms were both free-text and MESH headings and were combined with Boolean operators. Key search terms included the key words 'fall*' and 'older adults', 'seniors', 'preference', 'attitudes' and a wide range of technologies (Table 1). The MEDLINE search is available as supplementary material. The electronic searches are up to date at 01 April 2013.

2.2. *Inclusion/exclusion criteria*

All types of study designs were included, since most studies in this area of research are exploratory and there are few RCTs.

This review considers all studies that include older adults aged 50 and above. This is to reflect the attitudes and

experiences of younger older adults (the 'young old') as well as those past retirement age and older.

Studies had to include technologies related specifically to falls prevention, detection or monitoring. Technologies had to be designed to be directly used by older adults. For example, electronic patient records or staff training technologies were excluded. The studies included provide evidence on older adults' attitudes/experiences/feedback on these technologies/interventions. The included studies related to (i) direct experience of technologies both in a home environment and in focus group sessions, (ii) studies where potential technologies were discussed, but there was not necessarily any exposure to the technology as an intervention.

2.3. Selection process

The selection of papers was carried out in two stages. First, two reviewers independently looked at the abstracts identified by the searches and agreed whether they should be included or not. A total of 4044 papers were on inspection of title and/or abstract excluded as they were not relevant, or were duplicates. The large number of irrelevant hits was in part a function of the search system on COMPENDEX at the time. The second stage involved reviewing the full paper against the inclusion/exclusion criteria. Any studies which either reviewer were unsure about at stage one, were included in the second stage. A third reviewer independently assessed the paper against the inclusion/exclusion criteria if consensus could not be reached.

2.4. Data extraction and quality assessment

Once a study had been included, detailed study quality assessment was conducted. Methodological quality was assessed by two reviewers independently (H.H.H and E.B.). To assess methodological quality, we used the Joanna Briggs Institute (JBI) tool [16] for qualitative research as it has been assessed as the most coherent available [17]. We used the Quality Assessment Tool for Quantitative Studies by the Effective Public Health Practice Project (EPHPP) for quantitative studies [18] as recommended by Cochrane [19], which gives clear guidance for weak, moderate and strong quality ratings. As many quantitative studies included in the review used cross-sectional questionnaires we put more emphasis on response rate and the use of validated and reliable tools, considering the "Fitzpatrick" [20] criteria when answering the EPHPP tool. For mixed methods studies we used both tools, but gave priority to the dominant method. We also considered triangulation and whether a mixed methods approach led to richer data. If divergence between two independent reviewers' assessments could not be resolved by consensus, a third reviewer (A.H.) assessed the paper. The quantitative and qualitative studies are summarised in separate data extraction tables. Where there are mixed method studies they are summarised by the dominant method. These data were extracted directly into tables by two reviewers and then condensed (E.B and A.H.). They were then assessed by a third reviewer (H.H.H.) (see Tables 2 and 3) containing summarised data). Full data provided as supplementary material. Narrative synthesis was adopted, using a thematic analysis approach for this review following The Centre for Reviews and Dissemination guidance (CRD) [21], as outcomes are too diverse to combine. Due to diversity of outcomes it was not possible to pool results for meta-analysis.

3. Results

3.1. Article selection

We identified 76 potentially relevant papers, which were assessed against inclusion criteria, 21 studies were considered for quality review and were included (Fig. 1). The main reasons for rejection were that technology was not specifically related to falls, that it was not possible to differentiate the data that was specifically related to falls technology, or that data were not collected on attitudes or stakeholders views on technologies. Characteristics of the included studies are described in Tables 2 and 3 (summarised in this paper and available in full as supplementary material).

3.2. Methodological quality

The consensus assessment of study quality, using JBI and EPHPP tools, is presented in Tables 2 and 3. Comparison between studies was difficult as there was no uniformity in the scales used to collect data, or in types of ICT device used.

There was limited quantitative literature and no RCTs. Most studies included are observational cross-sectional studies [22–27]. There was also one Controlled Clinical Trial [4] and one cohort study (pre and post) [28]. All of these studies were of weak or moderate quality, even when mixed methods were used (supplementary materials Tables 2 and 3). There were two mixed methods studies which were pre-dominantly quantitative [27,28]. All quantitative studies used different approaches and very few used previously validated tools [22,28]; therefore, we were unable to make comparisons.

The remaining studies were qualitative (Table 3) or mixed methods but pre-dominantly qualitative [23,25,26,29]. The majority of the studies were of moderate quality (Table 3) with few studies that were of good quality meeting the majority of the criteria [30,31].

3.3. Findings

In general, inclusion criteria for study cohorts recruited relatively healthy, community-based older adults. Cohorts over represented women, and ethnicity was not stated for most studies.

3.4. Personal emergency alarms/home automation systems (including video monitoring as part of a system)

Two quantitative studies [4,22], three mixed methods studies [23,26,29] and three qualitative studies [32–34] looked at personal emergency alarms or wearable "falls detectors". We class these as 'body-fixed' (attached to the body e.g. by transparent film, neoprene belt) or 'body worn' systems (worn on or near the body e.g. trousers pocket). One quantitative study

Table 2 – Quantitative studies in order of technologies (see supplementary material for more details).

Authors, date, location	Study design	Sample	Technology type intervention
Brownsell and Hawley, 2004 UK [4]	Controlled Clinical Trial	N = 55: 21 in control (mean age 80); 34 in intervention (mean age = 78). Gender and Ethnicity not reported. More falls experienced and higher FES scores in intervention group.	Technology type: Personal emergency alarms/home automation systems (including video monitoring). Automatic fall detectors (Attendo, Tunstall or Tynetec). Intervention: Waist mounted fall detectors worn. Calls and self-reported experience recorded. Monitoring period = 17 weeks (SD 3.1).
Lai et al., 2010 Hong Kong [24]	Survey	N = 368 (mean age not stated). 243 female; 90 male. Ethnicity not stated. Convenience sample. Members of district elderly community centres aged >65.	Technology type: Personal emergency alarms/home automation systems (including video monitoring). Personal Emergency Link Service (PELS); home-based motion monitoring system (MMDevice); wearable vital signs monitoring system (VSMDevice). Intervention: Structured questionnaire on attitudes to telecare devices.
Heinbuchner et al., 2010 Germany [22]	Observational Cross-Sectional	N = 52:(mean age 81.4 years) 37 female, 15 male. Ethnicity not stated. Community dwelling: 75% lived alone; 25% with spouse, children or relatives. No severe cognitive impairment.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of a system). Intervention: Using PERS for at least 4 months. Measure level of satisfaction of personal emergency response systems (PERS) use. Wearing time of portable help button.
Silveira et al., 2013. Italy [27]	Observational Cross-Sectional. Mixed methods.	N = 11 (mean age = 77.2 years). 6 female, 5 male. Ethnicity not stated. 70+ years; able to walk independently no severe illness, impairment or condition.	Technology type: Portable computers/two-way communication devices (video conferencing). Active Lifestyle App on iPad. Intervention: Strength and balance exercises with virtual group. Two week strength and balance training.
Wu et al. 2006 USA [28]	Exploratory Mixed methods	N = 17 (mean age = 81) 13 female, 4 male. Ethnicity not stated. Over 65 years; access to broadband independent living, ambulant, 1+ fall in last year or fear of falling.	Technology type: Portable computers/two way communication devices (video conferencing). Intervention: 15 week group tele-exercise programme (GTEP) to deliver Tai Chi Quan to improve balance. Instructor and all participants could see/hear each other.

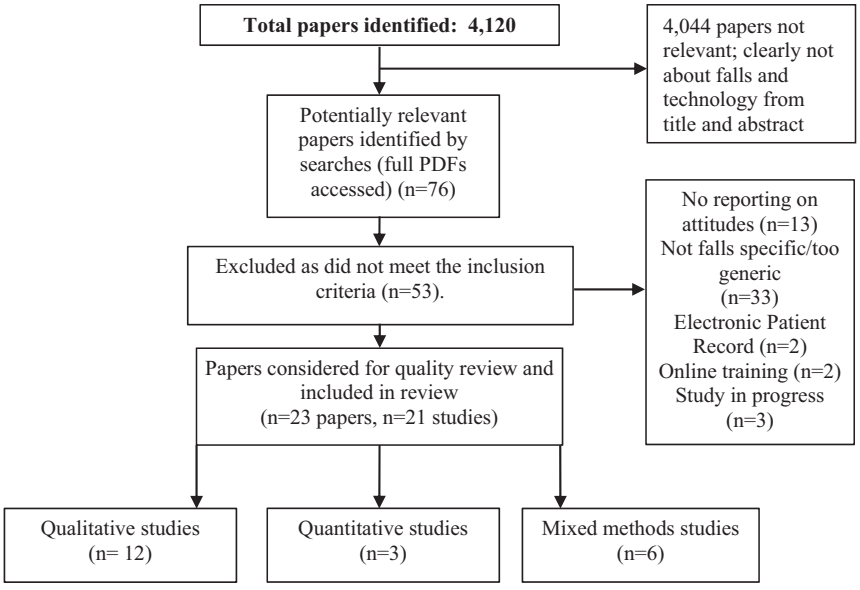


Fig. 1 – PRISMA diagram: flow of studies included and excluded in review.

Table 3 – Qualitative studies in order of technologies (see supplementary material for more details).

Authors, date, location	Study design	Sample	Technology type intervention
Bailey et al., 2011 Ireland [32]	Exploratory	N = 8 (mean age = 78). 5 female, 3 male. Ethnicity not stated. Purposeful sample, community dwelling, falls history.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of a system). Intervention: Four week intervention: daily activity log; weekly life-space diary (interests and pastimes) and pedometer for 1 week.
Blythe et al., 2005 UK [33]	Exploratory	N = 19 (mean age not stated). Gender not stated. Ethnicity not stated. Medical and care professionals (9 ambulance crew; 2 Occupational Therapists; 4 Mobile Wardens; 3 Age Concern staff; 1 Shop mobility manager); 4 older people.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of a system). Intervention: Questions about common call-outs and injury locations; equipment for independent living; client problems and daily routines.
Courtney et al., 2008 USA [35]	Exploratory	N = 14 (mean age not stated). Gender not stated. Ethnicity not stated. 11 in focus groups, 3 individual interviews. Community dwelling in independent living or assisted living facilities; 65 years or over.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Smart home technologies (bed sensor, kitchen sensor, motion sensor, fall detection sensor).
Demiris et al., 2004 USA [36]	Exploratory	N = 15 (mean age not stated). 8 female, 7 male. Ethnicity not stated. All participants over 65. Retirement community dwelling.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Questions about current and possible uses of technology re: health issues; privacy, confidentiality, usability and training. No actual trial of technology.
Demiris et al., 2008 USA [37]	Exploratory	N = 14 (mean age not stated). 9 female, 5 male. Ethnicity not stated. Participants over 65 years.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Questions about perceptions of usefulness of bed sensor, motion sensor, gait monitor, kitchen sensor, video-based fall detection sensor. Sensors shown and explained. No actual trial of technology.
Dorsten et al., 2009 USA [38]	Exploratory	N = 106 Residents: N = 12 (mean age = 85) 10F, 5 M; Residents family: N = 2 (mean age = 56). 1F, 1 M; Direct caregivers: N = 44 (mean age = 45). 41F, 1 M; Support staff: N = 20 (mean age = 50). 18F, 1 M; Maintenance staff: N = 17 (mean age = 48), 17 M; Administrators: N = 7 (mean age = 46). 5F, 2 M. Ethnicity not stated. Long-Term Condition Facilities	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Wide range of assistive technologies including surveillance and monitoring technologies.
Holzinger et al., 2010 Austria [23]	Exploratory. Mixed methods	N = not stated (mean age not stated). Gender not stated. Ethnicity not stated. All but 2 participants over 55 years; independently living in own homes.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: EMERGE Wrist Device prototype, to measure pulse rate and detect falls (impact and lack of movement). Data sent to wireless base station.

– Table 3 (*Continued*)

Authors, date, location	Study design	Sample	Technology type intervention
Horton, 2008 UK [34]	Observational	N = 35 (mean age = 78.2). 22 female, 13 male Ethnicity: Caucasian. 65+ years old; 2+ falls in last 6 months; community dwelling, living alone; registered with call centre.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Intervention Group: Fall detector, bed occupancy sensor, key safe. Control Group: Standard pendant alarm (already had). Monitoring period: mean 15.0 SD+ 4.2 weeks.
Hsin-Kai et al., 2012 Taiwan [39]	Exploratory	N = 30 (mean age = 59.2 years). 16 female, 14 male. Ethnicity not stated. Primary caregivers, caring for people with dementia.	Technology type: Personal emergency alarms/home automation systems (including video monitoring). Intervention: TMSS (telecare medical support system) installed for 6 months. T400 home units (Tunstall) connected to response centre, wireless pendants, fall detectors, various sensors. Sensors used to form decision rules, to detect anomalies.
Londei et al., 2009 Canada [25]	Exploratory sequential. Mixed methods	N = 25 (mean age = 79). 19 female, 6 male. Ethnicity not stated. 65+ years; 1+ falls in last year; no known cognitive impairment.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Video re: falls scenarios presented for feedback. No actual trial of technology.
Mathie et al., 2004 Australia [29]	Feasibility Mixed methods	N = 6 (mean age not stated, range 80-86). 4 female, 2 male. Ethnicity not stated.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Feasibility of waist mounted triaxial accelerometer (TA) (human movements and functional status. WiFi connection. Short routine undertaken daily, monitored for falls; metabolic energy measured. Worn for 2–3 months.
Milhailidis et al., 2010 Canada [26]	Exploratory Survey Mixed methods	N = 30 (mean age not recorded). 15 baby boomers (45–49). 10 female, 5 male. 15 older adults (70–74) 8 female, 7 male. Ethnicity not stated. Community dwelling in Toronto area.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Views on various technologies e.g., personal emergency response systems, fall detection systems, and sensor types e.g. switches, motion sensors, computer vision.
Steele et al., 2009. Australia [30]	Exploratory	N = 13 (mean age not stated). 7 female, 6 male. Ethnicity not stated. 65+ years, community dwelling, independently living.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Questions about current wireless sensor network (WSN) technologies and designs. No actual trial of technology, sample mote demonstrated
Van Hoof et al., 2011 Netherlands [31]	Exploratory Observational	N = 18 (mean age not stated). 14 female, 4 male Ethnicity not stated. 16 dwellings; 18 older adults. Receipt of home care visits, access to emergency response system; falls history, or wandering, or regular activation of alarm, or social isolation, or concerns about safety.	Technology type: Personal emergency alarms/home automation systems (including video monitoring as part of system). Intervention: Ambient intelligent system (sensors and set top boxes for videotelephony). Camera activated with alarm.

– Table 3 (*Continued*)

Authors, date, location	Study design	Sample	Technology type intervention
Broadbent et al., 2012 New Zealand [40]	Exploratory	N = 89 Residents: N = 32 (mean age = 82) Staff: N = 30 (mean age = 50) Relatives: N = 27 (mean age = 59) 70 female, 19 male. Ethnicity: 75 NZ Pakeha/Maori; 6 European; 6 other. Purposeful sample of retirement centre residents, staff, relatives. 21 in focus groups 90 completed questionnaires.	Technology type: Robotics Intervention: Healthcare Robots No actual trial of technology.
Doyle et al., 2010 Ireland [41]	Exploratory	Lab testing: N = 5 (mean age not stated). Gender not stated. Ethnicity not stated. Home testing: N = 7 (mean age not stated). Gender not stated. Ethnicity not stated. Prior fall(s).	Technology type: Games consoles. Intervention: BASE in home system to deliver Otago training. Flash application to play instructions; kinematic sensors to record movement. Testing in lab, then testing at home.
Uzor et al., 2012 UK (Scotland) [42]	Participatory	Workshop 1 N = 9 non-fallers (mean age = 67). 8 female, 1 male Ethnicity not stated. Workshop 2 N = 7 fallers (mean age = 73). 4 female, 3 male. Ethnicity not stated.	Technology type: Games consoles. Intervention: Workshops to influence design of interactive game to improve strength and balance.

[24], one mixed methods study [25] and seven qualitative studies [30,31,34–39] considered home automation systems. We class these as environmental systems, located in the environment chosen, rather than moving around with the person.

Control: Practically all studies talked about older adults having control over the technology and the situation. Control with regard to privacy from the technology was very important and mostly related to automation systems that included video imagery activated after a fall. Older adults said that they may accept a clear image being shown after they had fallen in more public sections of their home (e.g. living room), but not in the bedroom or bathroom [25,26]. This was particularly important for younger older adults, labelled as 'baby boomers' [26]. Blurred images, especially in more private sections of the house were seen as more acceptable [25,26]. This is echoed in the qualitative feedback from older adults [36–38]. In Londei et al.'s [25] study the 48% of participants who did not use the technology cited a breach of privacy as one of the reasons. Qualitative literature supports the argument that, for some older adults, cameras or visual surveillance are not acceptable in any format [30], although for some the trade-off for independence and security is enough to make the older person accept the technology [31].

Choice was an important element of control and also linked to decisions about the video cameras. Older adults stated that they liked systems or pendent alarms where they could cancel false alarms [4] or that were only used for 'emergency response' [30,37]. They wanted to manage the situation themselves. The technology had to be personalised and under user control [22,25,33]. Not being able to turn off false alarms was an important reason for lack of uptake of technology, or unhappiness with the technology. False alarms were seen as a nuisance both to the older person and others [23,25,31,34,39]. Participants in only one study [33] found that they liked the reassurance from calls when there was a false alarm.

Perceived need: Whether participants took up or maintained use of technologies related to whether they felt they were relevant to them. This was especially true for men [23,26]. The 48% of people who did not use the technology in Londei et al.'s [25] study said they were not old or ill enough to need it. They would take it up in the future if they lived alone, became more ill, or became at greater risk of falls. Participants in Demiris's [37] study said that the technology was for frailer older people. Factors influencing perception of need in the qualitative literature included self-perception of health, physical condition, mental and emotional condition, anticipatory living, the influence of family and friends, the influence of healthcare professionals, the physical environment, the technology type, and the perceived redundancy of the technology (they did not feel they needed it) [35].

Improved safety: Older people reported that they used the technology because they felt it improved their safety [4,22,25], particularly those technologies with real time monitoring or connected to a response system [4,39]. In Brownsell and Hawley's [4] study, 85% said it improved safety and in Londei et al.'s study [25], 84% of participants were favourable, or partially favourable, to technology for home security. The security of having a falls alarm or system in place, meant that older people felt they were able to 'take more risks' [34].

Outcomes: Independence and confidence. Brownsell and Hawley [4] reported that 58% of participants thought the technology improved independence; 72% felt more confident after using the technology and 90% reported that they were pleased

they had a fall detector. Independence and need were closely linked, with perceived importance being associated with increased tiredness, need for help and increased depressive symptoms [22]. Number of falls was not related to perceptions or receptivity to technology [25]. This is supported by qualitative research where older adults cite motivation to use technology as related to the importance of independence [31,33]. This is again related to control, having choice and control in life and home. The technology enables them to take risks to remain independent [33].

Usability/adaption to home. Usability, reliability and how the technology integrated into the home were all important to the adoption and long term use of the technology. In Holzinger et al.'s study [23], 42.8% of participants thought that the functions of a falls alarm system were useful (wrist worn device). In Brownsell and Hawley's study [4], 38% of participants had difficulty attaching or wearing the device. The need for a smaller help button; greater range of transmitter; higher durability; better adjustment and more comfort; effectiveness and ease of use were cited by participants as the most important aspects of technology [22]. The need for a light, unobtrusive device in different colours was cited as important in Holzinger et al.'s study [23]. Feedback indicated that buttons were difficult to use and not intuitive; that grey text on a grey screen was difficult to read, although the wrist device was more comfortable to wear than comparable devices [23].

Qualitative research supports these findings suggesting that technology needs to work with older adults' existing spaces and strategies and that older adults do not want the equipment to identify them as 'fallers' [32,37]. The technology needs to be reliable, simple to use and tailored to them [30,36,37]. Participants were found to reject technology, as they did not want their home to look like a hospital [33].

Cost. Concerns about cost and maintenance did not come up in quantitative studies but was cited by participants and stakeholders as a concern in qualitative studies [26,30,36–38]. Higher social economic class was found to relate to higher use [26].

3.5. *Portable computers/two way communication devices (video conferencing)*

Only two quantitative studies [27,28] looked at the use of portable computers [27] and communication systems [28]. Both systems were used to deliver evidence-based exercise programmes (Tai Chi and a strength and balance programme). We class these as portable systems as they can be moved within the environment, but are not body worn, or body fixed. *Convenience and commitment/motivation to continue*. Participants suggested that carrying out exercises using an iPad or teleconferencing in this way was convenient and, for the Tai Chi group, let them exercise as a group without issues of access to venues [28]. Participants who carried out exercises at home were eager to continue [27,28]. In the Silveira et al. [27] study, there was high intention to use the exercise application (Active Lifestyle App) on the iPad again and to recommend it to friends or family members (100%). Most of the participants did not feel motivated performing physical exercises in general, but felt motivated with the Active Lifestyle App (64%) because it was fun to follow the exercises with the App (91%). Most participants felt motivated when they saw the flower growing (used as an indication of user progress), or could monitor their progress towards their plan (both 91%).

Usability. Usability could have related to this high level of satisfaction, as participants in both studies needed little help using the technology after the first few weeks. Participants in both studies found the technology easy to use [27,28]. Participants reported that clear visual and audio instructions were satisfactory. They found having virtual access to a personal trainer at home allowed them to talk to the instructor about what worked and what did not [28]. Participants in the Silveira et al. [27] study were also happy with the delivery and support, expressing high perceived usefulness of the exercise videos. They found them helpful for guidance and adjustment of posture and reported that the weekly exercise calendars also let them check planned workout sessions.

Social motivation. Social motivation was an important element of the use of the technology and for carrying out exercises in both the Silveira et al. [27] and Wu et al. [28] studies. Silveira et al. [27] used specific social activities to promote the use of the technology and the majority of the participants felt motivated by being part of a training plan group (91%). They found that monitoring and comparing progress by the use of the flower was similarly effective (64%), as participants could compare their progress with others. Knowing that other people were doing the same exercises and facing similar challenges (e.g., pain, difficulty wearing the ankle weights) helped them to feel supported. The teleconferencing in the Wu et al. [28] study allowed participants to share experiences, opinions, and/or concerns, and made them feel supported.

3.6. *Robotics*

Only one exploratory qualitative study [40] looked at robotics, which we have classed as a portable system. Residents of a retirement village, their families and carers, were asked for feedback on healthcare robots that could offer a range of support. This included communication; help with ADLs; support to walk to places etc. Feedback from participants was similar to that relating to falls detectors and home automation systems, where participants said that video recording and cameras were unacceptable. Usability and reliability were important, with a demand for a clear screen and large buttons. There were concerns from families about the robot taking away human contact. Similarly, staff had concerns about it taking away their tasks. Overall, participants were more positive about the robots than families and staff, finding the robots useful for communication.

3.7. *Games consoles*

There were only two qualitative studies which looked at the use of game consoles. Both systems were specifically developed for the project under study and used body sensors to detect movement; they are therefore classed as body-fixed systems. Both of these studies delivered evidence-based exercises [41,42].

Feedback. In the Uzor et al. [42] study, the need to address motivation and chart progress was very important, with feedback being interactive. Different difficulty levels were

requested by participants and it was important to achieve higher scores and to improve on previous scores. Participants wanted to be able to set a variety of objectives, goals and environments in the games, to keep them motivated. Clear audio feedback was expressed as important in the Doyle et al. [41] study, particularly from the physiotherapists, as older people felt being monitored encouraged them to continue. The participants also wanted the game/exercise to be needs-led and allow them to have full control over the pace of the programme. Feedback made participants feel that they were achieving the health benefits they wanted (reduction in falls).

Usability. Usability was mentioned, including the need for prompts for navigation and large text and buttons. This echoed some of the feedback on other technologies [41]. Participants also did not want to see themselves on screen and preferred a matchstick man (icon/avatar) representing their movements.

4. Discussion

The literature related to technologies aimed at predicting, monitoring and preventing falls suggests that intrinsic factors related to older adults' attitudes around control [25,26,33,36–38,43], independence [4,31,33] and perceived need/requirements for safety [4,22,23,25,26,35,37] are very important for their motivation to use and continued use of technologies. Extrinsic factors such as usability [4,22,23,27,28,30,32,33,36,37,40,41], feedback gained [41,42] and, to a lesser extent cost [26,30,36–38], are very important elements that support these attitudes and perceptions. These findings are not dissimilar to previous research carried out with older adults around attitudes towards the uptake of falls prevention interventions and attitudes towards exercise intervention, which also cited attitudes, perceived need, identity, control and independence as important motivational factors [12,13,44,45].

4.1. *Recommendations*

Choice and control over technology use is important. Developers should ensure that technologies can be controlled by the user, particularly in relation to video imagery. Based on older people's preferences, camera/visual surveillance technologies should only be used in response to an alarm trigger. The options of using blurred or outline images may be acceptable. There should be a very clear indicator of when cameras are activated; so that participants can feel reassured their privacy is not being compromised. Given the differing results in studies included in this review, further research into the relationship between perceived breach of privacy and the potential benefits of technology is needed. Giving older adults the ability to cancel false alarms is particularly important both in terms of the older adult maintaining control and in reducing the nuisance to others. Follow up should be provided to ensure that the person does not require further falls assessment, as they may have cancelled an alarm call following a non-injurious fall.

Understanding and accepting the need for technologies is important. Technologies must be seen as useful in order for older adults to accept them. Potential benefits such as independence, increased safety, convenience, increased social opportunities and improvements in function and confidence should be promoted when encouraging older adults to accept technologies. Aesthetic design issues may be as important as the function in ensuring uptake and adherence to technology ("must have technologies"). Social opportunities and group cohesion are also important elements which could be considered as part of the development of any intervention. Regular real time feedback related to the individuals' needs and goals has been successful in achieving motivation to continue and should be included in technologies that promote physical activities.

Finally, usability and cost for individual users must be carefully considered. Technologies should integrate well into the home, be easy to operate and be reliable. Specific advice about large buttons, clear screens and both auditory and visual messages are important to meet the needs of all older adults. The eventual cost of technologies and whether they can be sustained long term by an individual will be important to long-term use.

4.2. *Limitations*

There are various limitations to the review. The methods adopted means that only studies which are published and are in English have been included. The majority of the evidence is exploratory, making it difficult to provide robust conclusions. However, randomised controlled trials are unlikely to give us details on older adults' attitudes and perspectives on falls technologies. Indeed our search strategy specifically targeted attitudes and perceptions of older people, but papers may have been missed if psychological findings about a technology were not core to the study objectives, but comprised a tangential finding [46]. Finally, these results can only be generalised to fairly healthy, community living older adults and there are problems in the representativeness of samples in the studies reported. Most samples were convenience sampling based rather than strictly population samples, hence generalisability of results to community-based older people in general and to higher risk groups in particular may be problematic.

4.3. *Conclusion*

Overall, there is a limitation in the conclusions that can be made from exploratory studies. However, it is reassuring to find that information we have already established about older adults' attitudes to general falls interventions and exercise interventions can be applied to falls technologies. We can also continue to learn from older adults' perceptions and attitudes towards general technologies use. Positive messages about the benefits of falls technologies for promoting healthy active ageing and independence are critical, as well as ensuring that the technologies are simple, reliable, effective and tailored to individual need.

What is clear is that different technologies used for falls prevention need to be considered in different ways. The technologies themselves need to be clearly described in research and older adults' attitudes towards different sorts of technologies must be clarified if specific recommendations are to be

Summary points

What was already known on the topic

- Falls are a serious issue for older adults and technologies can help prevent them but also help older adults deal with them.
- One of the main barriers to the successful use of these technologies is lack of adoption/adherence to their use.

What this study added to our knowledge

- Intrinsic factors related to older adults' attitudes around control, independence and perceived need/requirements for safety are important for their motivation to use and continue using technologies.
- Extrinsic factors such as usability, feedback gained and cost are important in supporting attitudes and perceptions.
- The positive benefits of technologies for falls prevention need to be emphasised, including how they can promote independence.
- Ensuring technologies are tailored to individual need, reliable and effective are essential for long-term use.

made. For example, attitudes towards surveillance or CCTV type technologies were very different to body worn sensors and alarms. This review highlights that we have a lack of detailed information about older adults' views on smartphone technologies and also other stakeholders' views on the use and adoption of falls technologies. There is very clearly need for more research in the area overall.

Author's contributions

Helen Hawley-Hague and Elisabeth Boulton managed the review process with input from Alex Hall, Chris Todd and Klaus Pfeiffer. Klaus Pfeiffer and Helen Hawley-Hague designed the search strategy. Helen Hawley-Hague and Elisabeth Boulton reviewed the papers with Alex Hall acting as 3rd reviewer. All authors critically revised the paper for publication.

Competing interests

The authors declare that they have no proprietary, financial, professional, or other personal competing interests of any nature or kind. Klaus Pfeiffer was a co-author of one of the papers included but was not involved in the inclusion or quality assessment of the paper.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ijmedinf.2014.03.002.

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